# Mobile Price Prediction using Machine Learning

## Feature Extraction & Analysis

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# Introduction

01

## Market Competition

Mobile phone market is highly competitive

02

## Price Factors

Prices depend on multiple features

03

## Project Aim

Aim: Predict mobile price using ML

04

## Benefit

Helps in better pricing strategy

# Objective

### Feature Identification

Identify important features affecting price

### Model Building

Build prediction model

### Feature Analysis

Analyze feature importance

### Business Insights

Provide business insights

# Dataset Description

### Mobile Pricing Dataset Overview

This dataset lists mobile phone models with key features and prices, highlighting market variations.

# Data Preprocessing

01

### Remove missing values

Removed missing values to ensure data completeness and accuracy.

02

### Clean numeric data

Cleaned numeric data (RAM, Battery, Camera) to standardize and correct values.

03

### Convert text to numbers

Converted text to numbers for better data processing and analysis.

Example:

"5000mAh → 5000"

04

### Remove unwanted columns

Removed unwanted columns to focus analysis on relevant data.

# Feature Extraction

01 **Correlation Heatmap**

Used correlation heatmap

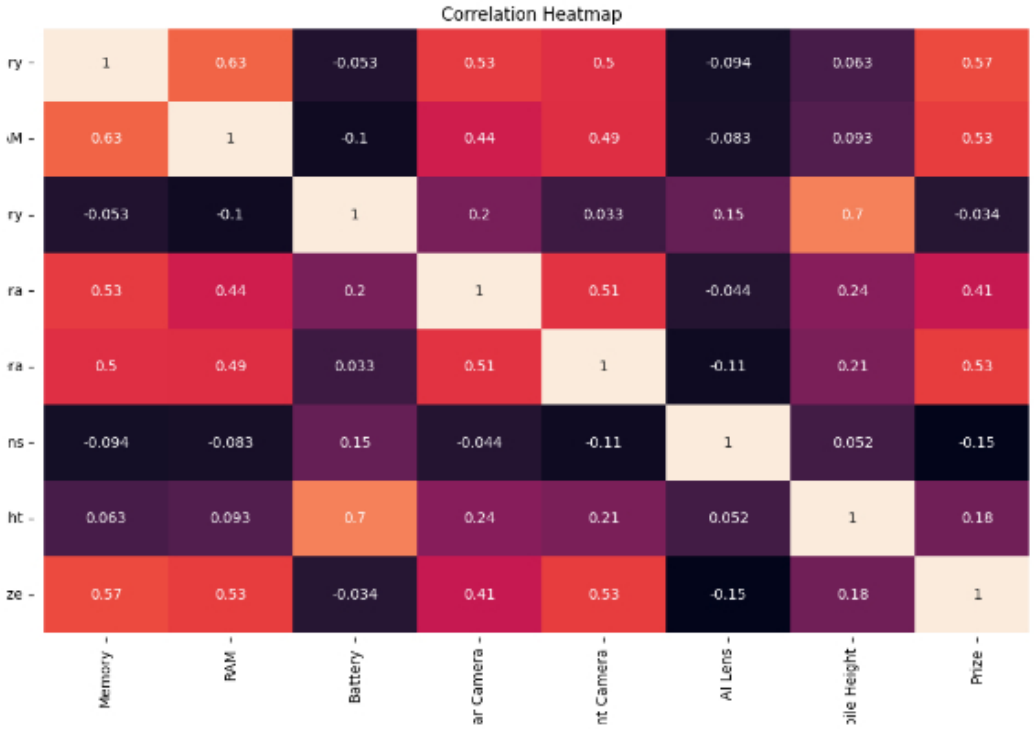
02 **Feature Identification**

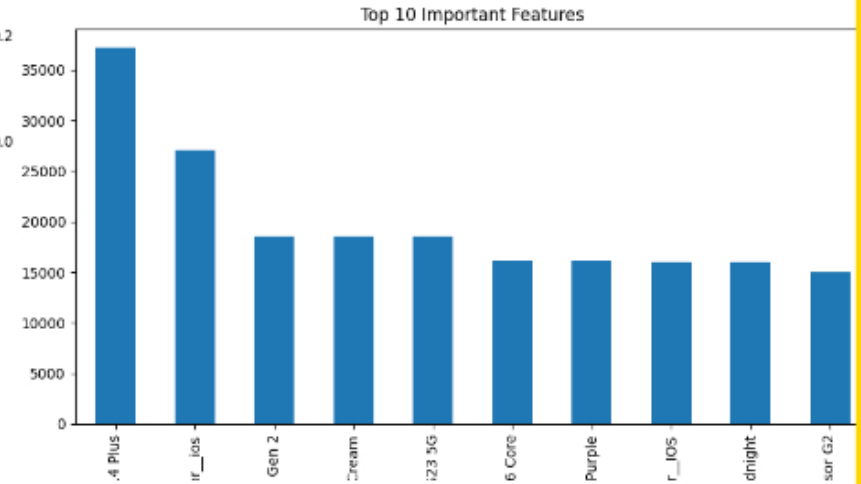
Identified important features

03 **Variable Selection**

Selected relevant variables

# Data Visualization





# Model Building

01

## Use Linear Regression

Used Linear Regression

02

## Split Data

Split data into training & testing

03

## Train Model

Model learns relationship between features and price

# Model Evaluation

**MAE (Mean Absolute Error)**

**2683.970**

**RMSE (Root Mean Squared Error)**

**6715.324**

**Accuracy (R8 Score)**

**0.71595**

# Feature Importance

### RAM Impact

RAM has strong impact

### Memory Influence

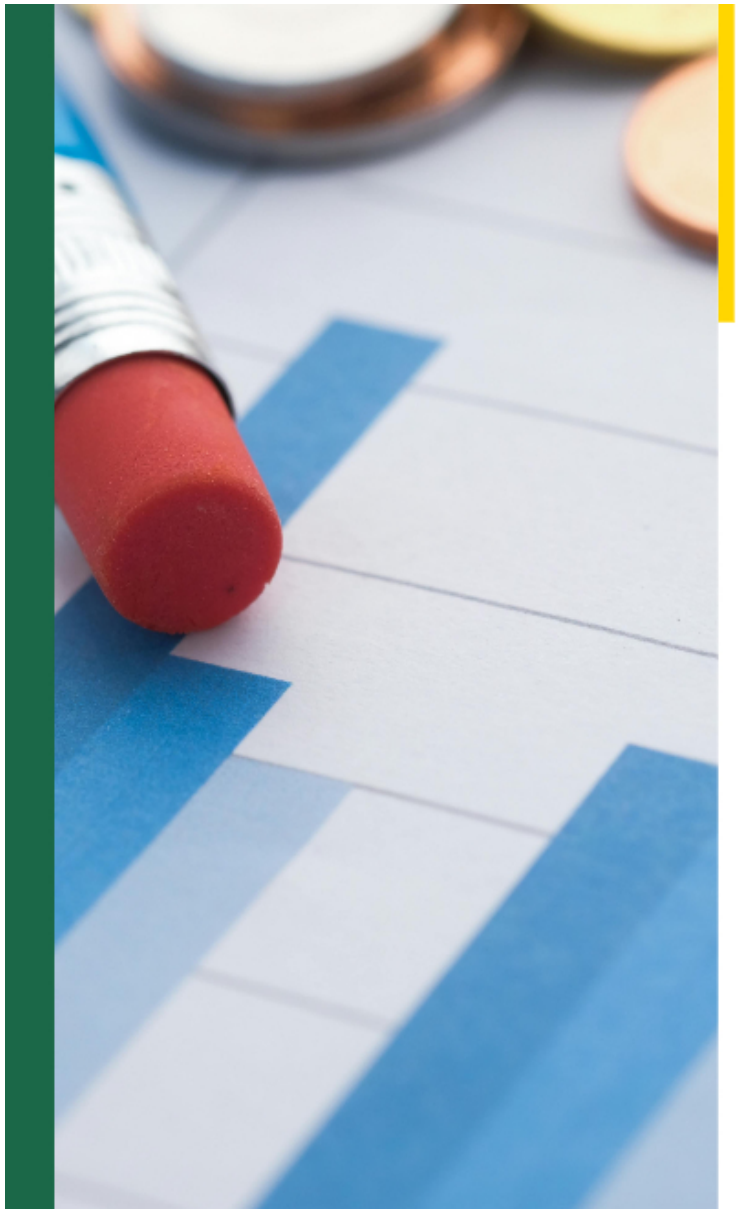
Memory affects price

### Battery & Camera

Battery & Camera also important

### Processor Role

Processor plays role





# Conclusion

### Price Prediction

ML model can predict mobile prices

### Key Features

Features like RAM & Memory are key

### Business Impact

Helps companies in pricing decisions

# Thank You!